

Brain PET imaging in obesity and food addiction: current evidence and hypothesis.



The ongoing epidemics of obesity is one main health concern of the present time. Overeating in some obese individuals shares similarities with the loss of control and compulsive behavior observed in drug-addicted subjects, suggesting that obesity may involve food addiction. Here, we review the contributions provided by the use of positron emission tomography to the current understanding of the cerebral control of obesity and food intake in humans. The available studies have shown that multiple areas in the brain are involved with the reward properties of food, such as prefrontal, orbitofrontal, somatosensory cortices, insula, thalamus, hypothalamus, amygdala, and others. This review summarizes the current evidence, supporting the concepts that i) regions involved in the somatosensory response to food sight, taste, and smell are activated by palatable foods and may be hyperresponsive in obese individuals, ii) areas controlling executive drive seem to overreact to the anticipation of pleasure during cue exposure, and iii) those involved in cognitive control and inhibitory behavior may be resistant to the perception of reward after food exposure in obese subjects. All of these features may stimulate, for different reasons, ingestion of highly palatable and energy-rich foods. Though these same regions are similarly involved in drug abusers and game-addicted individuals, any direct resemblance may be an oversimplification, especially as the heterogeneities between studies and the prevalent exclusion of sensitive groups still limit a coherent interpretation of the findings. Further work is required to comprehensively tackle the multifaceted phenotype of obesity and identify the role of food dependency in its pathophysiology. Copyright © 2012 S. Karger GmbH, Freiburg.